When do zero-shot single-cell foundation model embeddings beat classical representations for immune cell-type annotation in tumour microenvironment data?

All four hypotheses are resolved. Figures F1–F5 accompany this report; every number below is reproduced from the tables in `results/`.

1. What this study is for

This is a decision rule, not a demonstration. The question is not whether foundation models *can* produce useful cell embeddings — they can — but under which conditions a practitioner annotating immune cells in tumour data should prefer a zero-shot foundation-model embedding over a well-tuned classical representation, and under which conditions they should not.

Three axes were varied: **label budget** (5, 10, 25, 50, 100, or all labelled cells per class), **distribution shift** (within-dataset, held-out donor, held-out dataset), and **cell-type rarity**. Two foundation models (Geneformer V2-104M, scGPT-human) were compared against three classical baselines (HVG+PCA, scVI, Harmony) over **13 datasets spanning 8 cancer types and 229,801 cells**, using an identical classifier head throughout so the comparison isolates the representation.

The unit of replication is the dataset (n=13), never the seed. Seeds are averaged to a dataset mean before any test.

2. Limitations — read before the results

These are placed before the findings deliberately. Three of them bias toward conclusions this study reaches, and a reader who meets them afterwards has already formed an impression the caveats should have shaped.

2.1 Caveats that flatter the classical arms

scVI receives depth-normalised pseudo-counts, not raw UMI counts. TISCH2 distributes `log1p(CP10K)`; measured, per-cell `sum(expm1(x))` is exactly 10,000. scVI's negative-binomial likelihood requires integer counts. True counts are exactly recoverable for 9 of 13 datasets but not for the 4 plate-based ones, and using true counts only where available would inject a per-dataset artefact into a contrast whose unit of replication *is* the dataset. `round(expm1(x))` was therefore applied uniformly. Consequence: library size is ~10^4 by construction, so scVI's library-size latent is uninformative.

scVI's 60-epoch budget was never reached by early stopping. All 75 fits ran the full cap, so the budget — not convergence — terminated training.

Together these mean **scVI's numbers are a lower bound for two independent reasons**, and this study does not establish scVI's ceiling.

Harmony is fit transductively — a declared deviation from the leakage control. harmonypy 2.0 exposes no out-of-sample projection. Harmony therefore sees test-cell *expression* (never labels) while every other arm does not. This **advantages a baseline**, biasing toward "classical methods win", which is this study's own expected direction. It must not be read as evidence for H1. PCA remains train-only; only the correction step sees test coordinates.

2.2 Scope caveats

- **One checkpoint per model family.** Results are about *these two checkpoints*, not about "foundation models" as a class.
- **Zero-shot only.** No fine-tuning. Fine-tuned performance is out of scope and may differ substantially.
- **n=13 datasets.** Adequate for the large effects reported; underpowered for weak associations, which is why H4 is reported as "no evidence for" rather than "evidence against".
- **Pretraining-overlap confound, partially bounded.** BRCA_GSE176078 is a suspected member of scGPT's CELLxGENE-derived pretraining corpus, retained with a pre-specified sensitivity analysis. **That analysis is reported in §3.8: excluding the dataset changes no conclusion** — zero sign flips and zero significance changes across the 36 primary contrasts, largest shift 0.0095, and the flagged dataset is not an outlier in the direction memorisation would predict (max |z| = 1.29 across 36 cells).
- **Two post-hoc design changes**, both declared: extra seeds under leave-dataset-out at budgets 5, 10, 25 and 50 (5→20 — four budgets, not the single cell an earlier draft of this bullet implied), and reduced seeds at the unrestricted budget (5→2, justified by the measurement that the seed does not change the training data there). **The first is audited in §3.9: recomputing the affected contrasts on the original 5 seeds changes nothing** — zero sign flips, zero significance

changes, largest shift 0.0076 — so the boost did not produce the H2 conclusion. It also bought almost no precision (CI-width ratio 0.99), because the replication unit is the dataset, not the seed.

- **Gene-space coverage varies by dataset** (Geneformer vocabulary coverage 0.560–0.969), a per-dataset caveat rather than a property of the method.
- **The ultra-rare regime is untested.** The <1% prevalence stratum contained only 12 (dataset, cell type) instances across the whole corpus — too few to test. Conclusions about rarity extend down to ~1% prevalence and no further.
- **The mechanism analysis (§3.7) is descriptive.** It correlates across five representations, which are not independent replicates, on a fixed subsample. It motivates the next experiment; it does not establish causation.

3. Results

All contrasts are paired across 13 datasets, with bootstrap 95% CIs, Cohen's dz, Wilcoxon signed-rank p, and BH-FDR correction across the 108-contrast family. **49 of 108 contrasts are significant at FDR 0.05; none of them is flagged negligible** ($|\Delta| < 0.02$).

3.1 H1 — in-distribution, at restricted label budgets: no advantage

At every restricted budget from 5 to 100 labelled cells per class, within dataset, neither foundation model differs detectably from HVG+PCA: all ten contrasts return "no detectable difference" and **none is significant at FDR 0.05** (p range 0.150–1.000).

Point estimates run from –0.059 to +0.001 — where they lean at all, they lean *against* the foundation models. Two details matter for reading this honestly rather than as a flat null:

- The largest deviation is **scGPT at 5 labels per class, Δ = –0.059 [–0.115, –0.010]**, winning on only 3 of 13 datasets. Its bootstrap CI does *not* cross zero, yet its FDR-corrected p is 0.150. The CI is on the mean difference while the pre-specified primary test is Wilcoxon signed-rank on ranks, so the two can disagree; the pre-specified test governs. The correct reading is a *suggestive but unconfirmed* disadvantage for scGPT at the smallest budget — not an established one, and not a flat null either.
- Excluding that cell, the remaining eight contrasts (budgets 10–100) span –0.017 to +0.001, every CI crosses zero, and every FDR p > 0.64.

This confirms the study's stated expectation, and it is the finding with the most direct practical consequence.

3.2 The exception: unrestricted labels, in distribution

Given *all* available labels within dataset, both models beat HVG+PCA on **13 of 13 datasets**: Geneformer +0.051 [+0.040, +0.063], scGPT +0.052 [+0.039, +0.063], both FDR p = 0.0018, dz ≈ +2.3. This contradicts the stated expectation and is reported as such.

3.3 The strongest result: the advantage reverses under dataset shift

Under leave-dataset-out, HVG+PCA wins, and **the gap widens as labels increase**:

Budget	Geneformer – HVG+PCA	FDR p	Datasets won
5	–0.053	0.255	5/13
10	–0.064	0.069	4/13
25	–0.076	0.006	1/13
50	–0.083	0.002	1/13
100	–0.100	0.002	0/13
all	–0.107	0.002	0/13

scGPT shows the same pattern, more weakly (–0.083 at unrestricted budget, FDR p = 0.004, 2/13). **This is the opposite of the "foundation models generalise better" prior.**

3.4 H2 — against the other mandatory baselines

Direction depends on the same axis. In distribution at unrestricted labels, both scFMs beat scVI (+0.092) and Harmony (+0.137). Under dataset shift, both lose to scVI (–0.086 to –0.111) and to Harmony (–0.071 to –0.096) — despite scVI operating under two handicaps and Harmony under a declared advantage.

3.5 H4 — gene-space overlap: a null

No association between train–test gene-space overlap and the scFM advantage, across overlap spanning 0.538–0.946: Spearman ρ = –0.34 (p=0.25) and –0.13 (p=0.67) at 10 labels; +0.02 (p=0.96) and –0.09 (p=0.78) at all labels. Every point estimate is at or below zero — if anything, opposite to the prediction. Underpowered at n=13 to exclude a weak association.

3.6 H3 — rarity: no rescue, and no trend

Per-class F1 was recovered by refitting at the already-selected regularisation (no inner CV, so no selection on test). Classes were binned by pre-specified prevalence strata; the paired unit is the (dataset, cell type) pair.

Rarity does not change the direction of any result. In distribution at unrestricted labels, the scFM gain is present in *every* stratum and is largest in the middle one (Geneformer +0.057 uncommon vs +0.035 rare, +0.037 common). Under dataset shift the penalty is present in every stratum too, and at unrestricted labels it is *largest for the rarest testable classes* (Geneformer –0.168, scGPT –0.175 in the 1–5% stratum, both FDR p < 0.005) — the opposite of the hypothesis that foundation models earn their keep on rare populations.

Tested continuously rather than binned, the effect does not trend with prevalence in 11 of 12 cells (|ρ| <= 0.14, p > 0.15). The single exception is Geneformer under dataset shift at 10 labels (ρ = –0.26, p = 0.010), which is uncorrected for the 12 tests and should not be read as an established gradient.

The <1% stratum had only 12 class-instances and was not tested. **This study therefore says nothing about genuinely ultra-rare populations**, which is where a rarity benefit would most plausibly live.

3.7 Mechanism: the embeddings encode dataset identity

The reversal in §3.3 has a measurable candidate explanation. Computing silhouette width twice on the same cells — once labelled by cell type (the signal) and once by dataset (the nuisance):

Representation	Cell type	Dataset
Geneformer	+0.017	**+0.047**
scGPT	+0.036	+0.026
HVG+PCA	+0.020	–0.010
scVI	**+0.059**	–0.035
Harmony	–0.001	–0.053

Geneformer separates datasets more strongly than it separates cell types. Both foundation models have positive dataset silhouette; all three classical representations have negative. Across the five representations, dataset silhouette orders perfectly with the macro-F1 lost going from within-dataset to held-out-dataset (Spearman ρ = +1.00, exact permutation p = 0.017).

This is consistent with the frozen embedding encoding batch structure a linear head cannot discount, and it is *descriptive*: n = 5 representations are not independent replicates, the silhouette is computed on a fixed subsample, and correlation across five methods cannot establish causation. It motivates the next experiment rather than concluding this one.

3.8 Robustness check 1 — pretraining-overlap sensitivity

`results/pretraining_overlap_audit.md` pre-specified that every primary contrast be recomputed with `BRCA_GSE176078` excluded. **That analysis was specified before any result was known but was never executed**; it is run here and reported in full. Nothing was re-tuned and no pre-specified analysis was altered: the same frozen aggregate → paired-contrast → BH-FDR path is used, and the only difference is which datasets enter it. The n=13 leg reproduces the shipped T3 exactly on all 108 contrasts (max difference 1e-16), which is asserted rather than assumed.

No conclusion changes. Across the 36 primary contrasts against HVG+PCA there are **zero sign flips and zero significance changes**; the largest shift in any point estimate is 0.0095, below the pre-specified negligibility floor of 0.02. The headline contrasts:

Contrast	n=13	n=12 (excl. flagged)
Within dataset, all labels, Geneformer	+0.0513 [+0.0400, +0.0632], p=0.0018	+0.0533 [+0.0415, +0.0653], p=0.0035
Within dataset, all labels, scGPT	+0.0516 [+0.0392, +0.0631], p=0.0018	+0.0507 [+0.0377, +0.0635], p=0.0035
Held-out dataset, all labels, Geneformer	–0.1071 [–0.1988, –0.0507], p=0.0018	–0.1086 [–0.2056, –0.0470], p=0.0035
Held-out dataset, all labels, scGPT	–0.0828 [–0.1354, –0.0411], p=0.0043	–0.0857 [–0.1417, –0.0408], p=0.0085

Across the full 108-contrast family (including scVI and Harmony comparators) there is **1 sign flip and 3 significance changes**, all reported rather than suppressed. None is a primary contrast and all four sit on effects at or below the negligibility floor or just at the FDR boundary: `scgpt − harmony` at held-out-dataset/10 labels moves +0.0028 → –0.0008 (both $p > 0.87$, i.e. a sign flip on a non-effect), and three comparator contrasts cross from p≈0.047 to p≈0.07 — losing significance purely because dropping a dataset costs one degree of freedom on n=13.

H3 (rarity) is unaffected too. Recomputed over the per-class table, 36 rows: 1 sign flip and 1 significance change, both in the 5–20% stratum at 10 labels on effects of $|\Delta| <= 0.012$, and both far below the negligibility floor. Every budget-all stratum — where the H3 conclusion lives — retains its sign and its significance, including the rarest-classes result under dataset shift (Geneformer –0.168 → –0.177, scGPT –0.175 → –0.187, both still $p < 0.003$).

The audit's second pre-specified check also returns nothing. It asked whether the scFM margin is unusually large *on the flagged dataset* relative to its margin elsewhere, which would itself be evidence of memorisation. Across 36 scheme x budget x model cells the largest standardised deviation is $|z| = 1.29$ and **no cell exceeds |z| = 2**. The flagged dataset is not an outlier in the direction memorisation would predict.

Tables: `results/T10_pretraining_sensitivity.csv`, `results/T11_memorisation_check.csv`, `results/T13_h3_pretraining_sensitivity.csv`.

3.9 Robustness check 2 — does the post-hoc seed increase carry H2?

Seeds were raised from 5 to 20 after the design was fixed, under leave-dataset-out. That is the condition H2 lives in, so a reader may reasonably ask whether the added replication produced the conclusion rather than merely measuring it. The affected contrasts are recomputed here on **seeds 0–4 only** — the original pre-boost set — beside the 20-seed values.

One correction to §2.2 while reporting this: the boost was applied to **four budgets (5, 10, 25 and 50)**, not to a single low-budget cell. All four are included below.

The conclusion is unchanged. Across all 24 affected contrasts: **zero sign flips, zero significance changes**, largest shift in any point estimate 0.0076 — again below the negligibility floor. The H2 comparisons against scVI and Harmony hold their direction and their significance at 5 seeds.

The more interesting result is *why* it makes so little difference. Median CI width is **0.0819 at 20 seeds versus 0.0831 at 5** — a ratio of 0.99, i.e. the extra 15 seeds bought essentially no precision. That is exactly what the design's own replication rule predicts: the unit of replication is the dataset (n=13), so the interval is set by between-dataset variance, which added seeds cannot reduce. The seed boost lowered within-dataset resampling noise, which was never the binding term.

So the post-hoc change was close to inert. It could not have manufactured the H2 result, and — stated plainly because it cuts against the decision to make it — it also did not deliver the power increase it was intended to deliver. The underpowered-cell concern that motivated it is better addressed by more datasets, not more seeds.

Table: `results/T12_seed_sensitivity.csv`.

4. Decision rule

On this evidence, for immune cell-type annotation in TME data:

1. **Annotating within a dataset you already have labels for, on a restricted budget** — use HVG+PCA. There is no measurable benefit from a zero-shot foundation-model embedding, and the classical pipeline is cheaper, faster, and interpretable.
2. **Annotating within a dataset with abundant labels** — a foundation-model embedding gives a real but modest gain (~+0.05 macro-F1, consistent across all 13 datasets).
3. **Transferring to an unseen dataset** — use HVG+PCA. The foundation-model embeddings are measurably worse here, and increasingly so as labels grow.
4. **Do not choose based on gene-space overlap.** It carries no signal.

5. Downstream implications

The reversal in (3) is the result with consequences beyond this benchmark, and §3.7 supplies direct evidence for why it happens: **both foundation-model embeddings carry positive dataset silhouette while all three classical representations carry negative**, and dataset silhouette orders perfectly with the transfer penalty across the five methods. The frozen embedding appears to encode dataset-of-origin structure that a linear head cannot discount, and more labels in a shifted domain help the classical pipeline more than they help a representation whose axes partly encode the wrong thing.

That is a candidate mechanism, not a demonstrated one — five representations are not five independent replicates. The natural next experiments follow directly: (a) test whether an explicit batch-correction step applied *to the foundation-model embedding* recovers the transfer performance, and (b) test whether fine-tuning, which is out of scope here, removes the dataset structure that zero-shot embedding retains. Either result would sharpen the decision rule from "do not use these embeddings for transfer" to "use them for transfer only after correction".

The rarity result adds a second implication: §3.6 rules out the most common defence of scFM embeddings — that they earn their keep on rare populations. Under dataset shift the penalty was *largest* for the rarest testable classes. The genuinely ultra-rare regime (<1% prevalence) remains untested here and is the one place that defence could still survive.

For practitioners, the operational reading is that **the deployment scenario determines the answer**, and the scenario where foundation models are most often advocated — transfer to new data — is the one where they performed worst here.

6. Reproducibility

Environment lockfile, split indices, run manifest and DECISIONS log are in the repository. The calibration partition of the three-way split was written and never read by this project; it is reserved for downstream conformal-prediction work. Analysis code was frozen and hash-recorded before the full-grid statistics were run.

Compute: a single NVIDIA T4 instance with 4 vCPU and 16 GiB RAM.

Figures

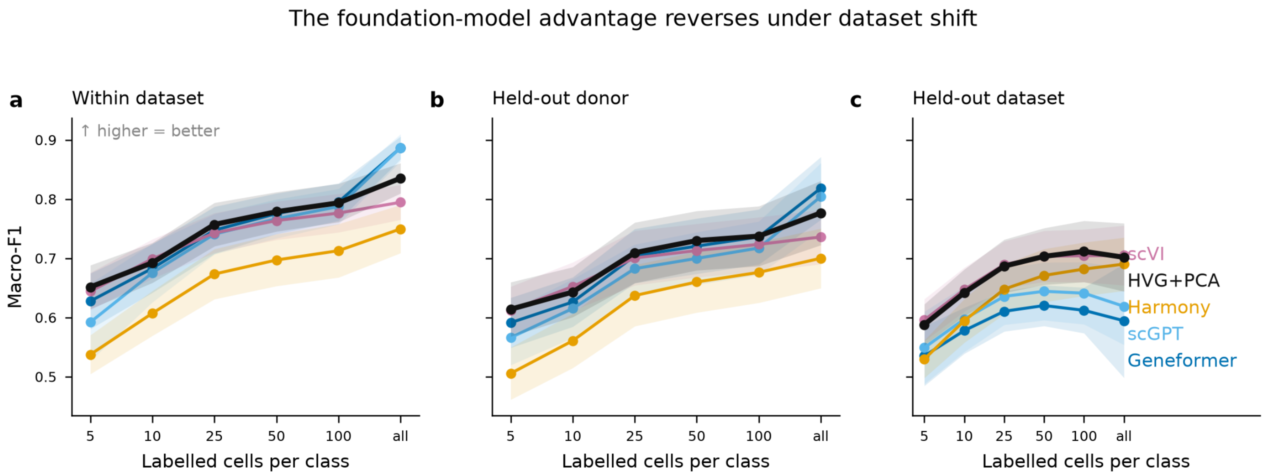

Figure 1. Macro-F1 against label budget under three shift regimes. Mean over 13 datasets; bands are 95% CI.

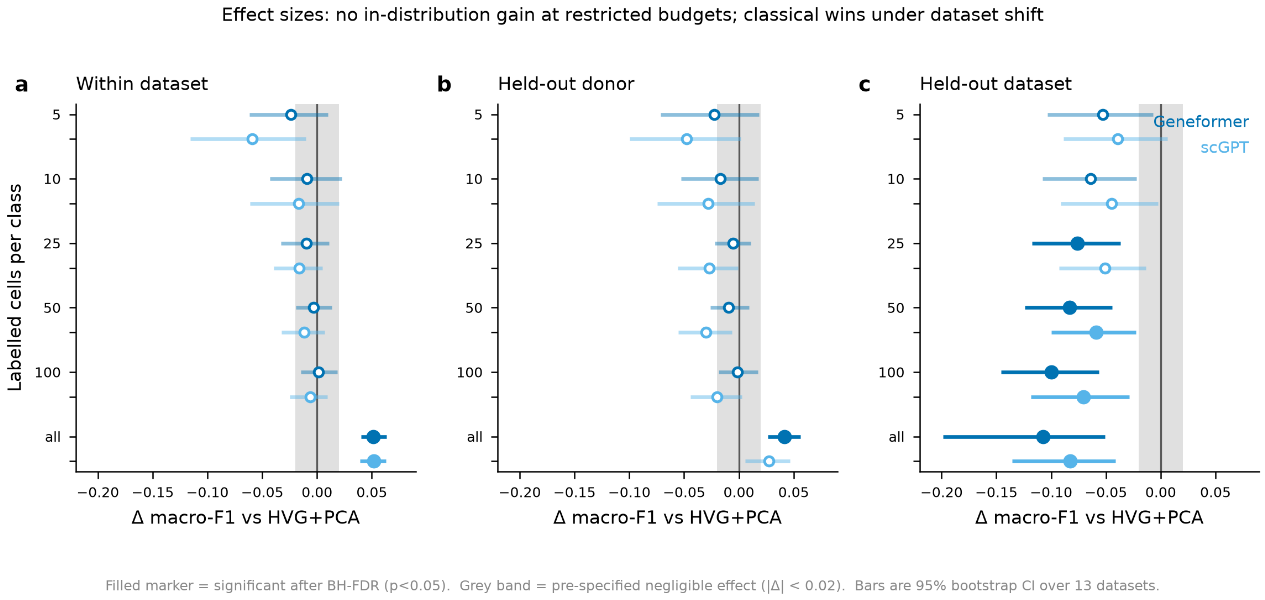

Figure 4. Paired effect sizes versus HVG+PCA with bootstrap 95% CI. Filled = significant after BH-FDR; grey band = pre-specified negligible effect.

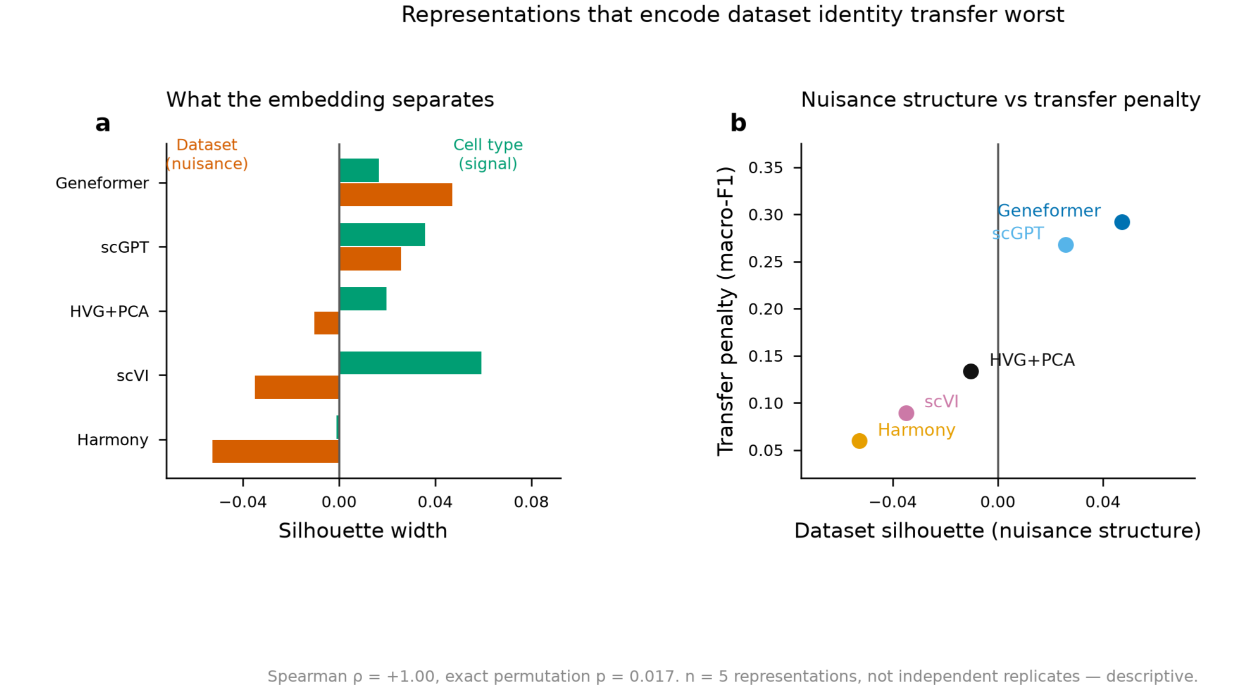

Figure 2. Signal versus nuisance structure in each representation, and its relation to the transfer penalty. Descriptive: n=5 representations.

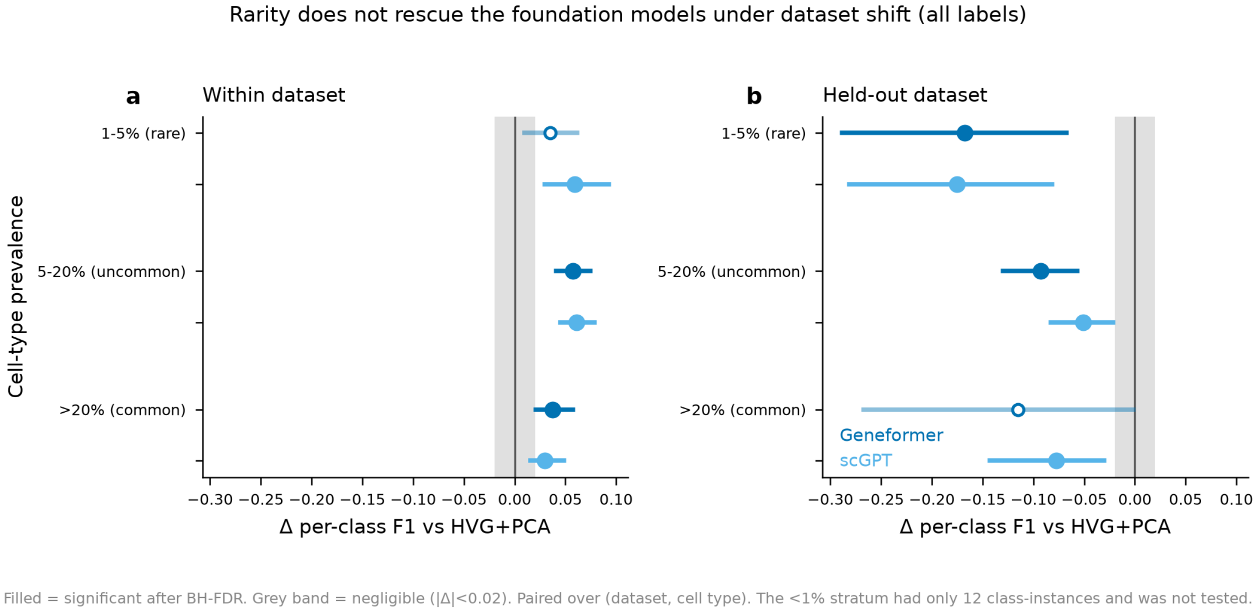

Figure 3. Per-class effect sizes by cell-type prevalence stratum, at unrestricted labels.

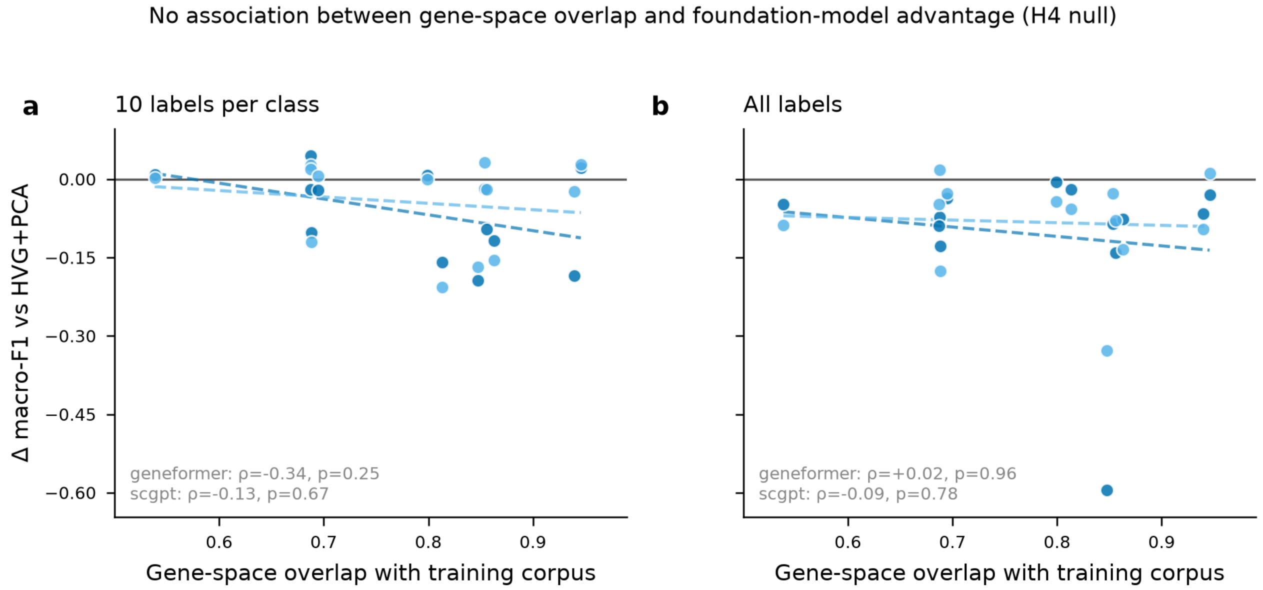

Figure 5. Foundation-model advantage against train-test gene-space overlap (H4, null).